TIC 403329469

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       30.939496 d  (ALIASED, see below)
    -> corrected period: 61.878992 d
  Epoch (t0)   2838.1348 BTJD
  Duration     0.5000 d
  Depth        0.051 %  (511 ppm)
  SNR          19.2

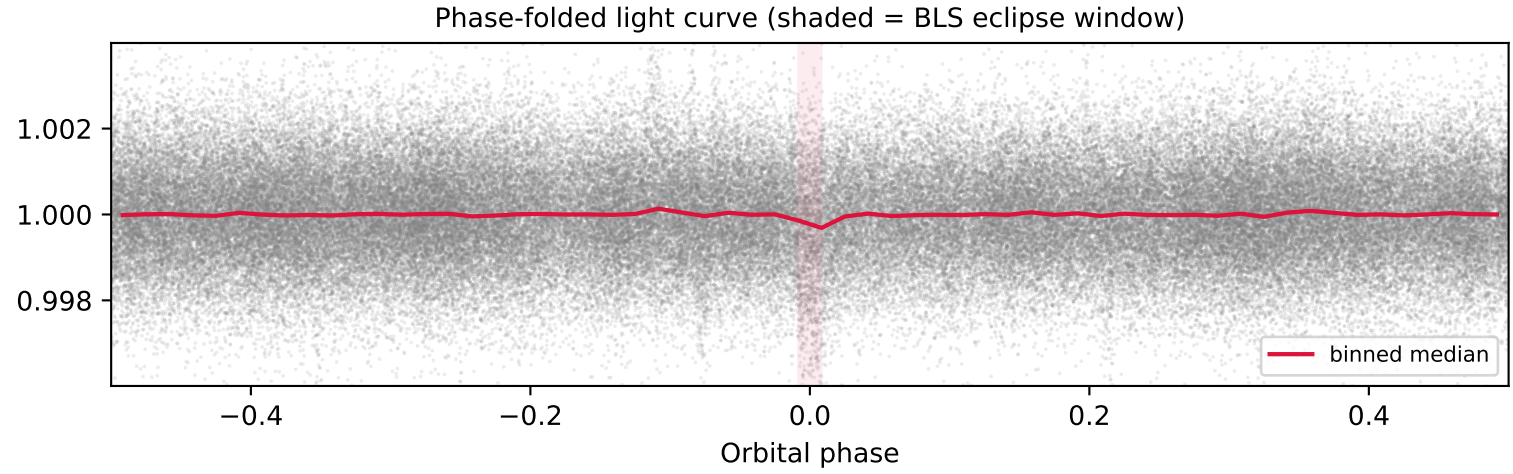
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     8  (gate >= 3)
  Robust depth ratio    0.69  (gate >= 0.50)
  Robust depth sigma    10.4  (gate >= 7.0)
  Odd/even z-score      20.32  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

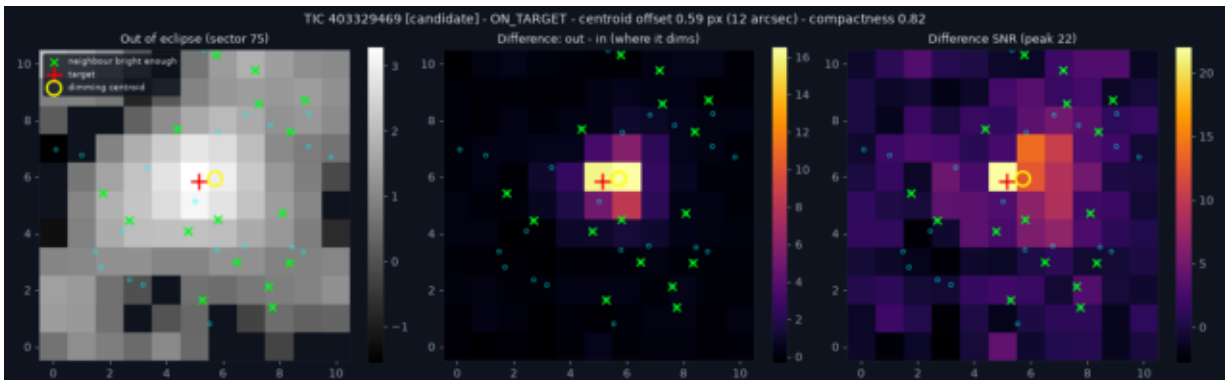
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 75    light-curve dip in this sector: 9.9 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   22.05   (gate >= 5.0)      Compactness   0.82   (gate >= 0.50)
  Dimming centroid offset     0.59 px (12 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 403333340
    1.45 px away, dTmag +7.78 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 20.32 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 61.878992 d, in any write-up.